

Identifying Variables — Set B — Answer Key

Level 3 · Ages 11-14 · Answers in bold. Accept any answer that means the same.

Independent variable — the factor you change, and the values you test.

Dependent variable — the factor you measure, with its units.

Controlled variables — factors kept the same in every trial.

Control group — a separate no-treatment trial used for comparison. If there isn't one, write "none".

- ① A student drops one fizzing vitamin tablet into 200 mL of water at 10 °C, 25 °C and 40 °C, using a new tablet of the same kind each time and never stirring. She times how long each takes to stop fizzing. (Do not drink the solution.)

Independent variable (and values):

Water temperature: 10, 25 and 40 °C.

Dependent variable (and units): **Time to stop fizzing, s (accept: time to dissolve).**

Three controlled variables: **200 mL of water; same kind of tablet; no stirring.**

Control group: **None (accept a justified baseline).**

- ② Parachutes made from the same paper, with canopy areas of 100, 200, 300 and 400 cm², each carry the same paper-clip load. Each is dropped indoors from 2 m, three times, and the mean time to reach the floor is recorded.

Independent variable (and values):

Canopy area: 100, 200, 300 and 400 cm².

Dependent variable (and units): **Mean time to reach the floor (s).**

Three controlled variables: **Any three: drop height 2 m; same load; same paper; indoors (no wind).**

Control group: **None (accept a justified baseline).**

- ③ A piece of pondweed in water is lit by a lamp placed 10, 20, 30 and 40 cm away. The water is kept at 25 °C. At each distance the gas bubbles (which contain oxygen) given off in one minute are counted.

Independent variable (and values):

Distance of the lamp: 10, 20, 30 and 40 cm (light intensity).

Dependent variable (and units): **Number of bubbles given off per minute.**

Three controlled variables: **Any three: same piece of pondweed; water at 25 °C; same lamp; one-minute count.**

Control group: **None (accept: pondweed kept in the dark, as a baseline).**